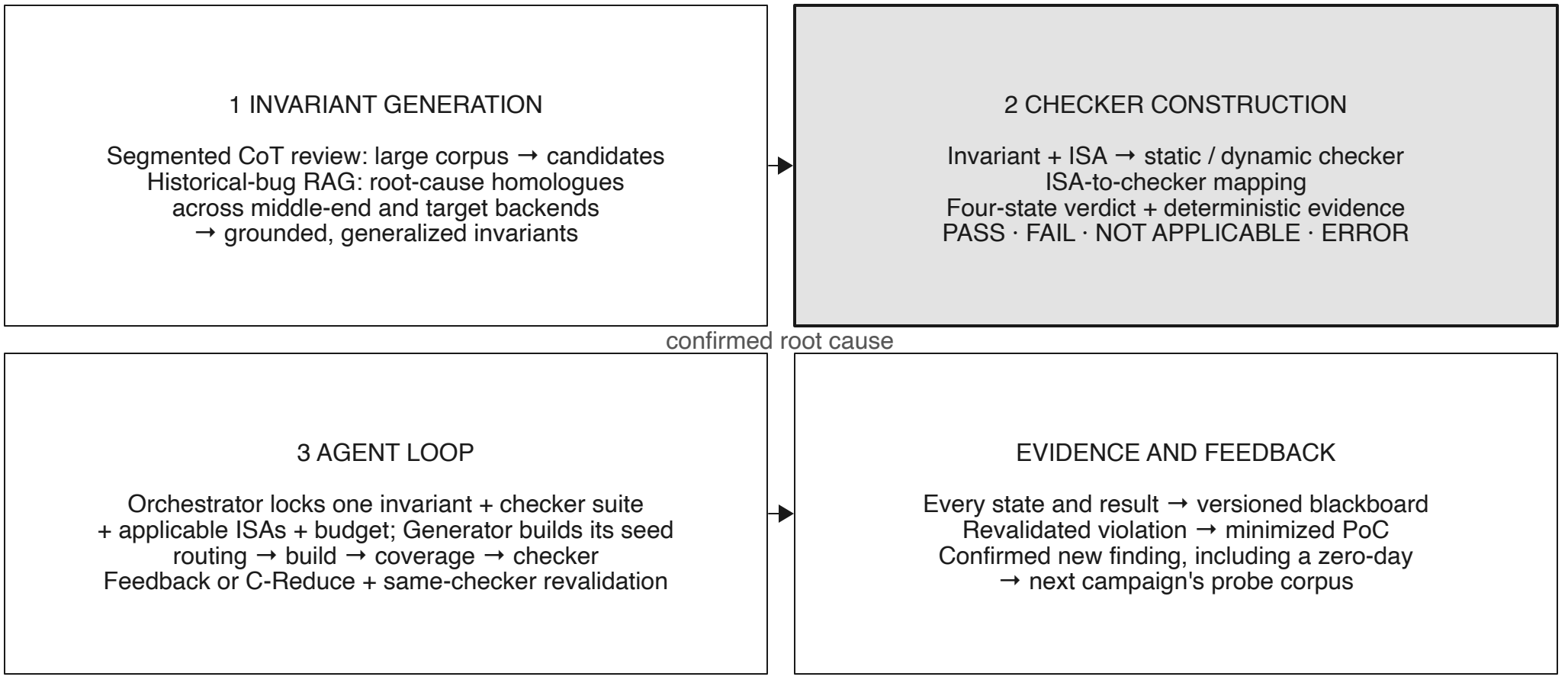
1 INVARIANT GENERATION
Segmented CoT review: large corpus → candidates
Historical-bug RAG: root-cause homologues
across middle-end and target backends
→ grounded, generalized invariants
2 CHECKER CONSTRUCTION
Invariant + ISA → static / dynamic checker
ISA-to-checker mapping
Four-state verdict + deterministic evidence
PASS · FAIL · NOT APPLICABLE · ERROR
confirmed root cause
3 AGENT LOOP
Orchestrator locks one invariant + checker suite
+ applicable ISAs + budget; Generator builds its seed
routing → build → coverage → checker
Feedback or C-Reduce + same-checker revalidation
EVIDENCE AND FEEDBACK
Every state and result → versioned blackboard
Revalidated violation → minimized PoC
Confirmed new finding, including a zero-day
→ next campaign's probe corpus